

MAT1002 Lecture 21,

Outline

- Change of variable (triple integrals)
- Spherical coordinates
- Line integrals of real-valued functions
- Vector fields
- Line integrals of vector fields

Change of Variable (Triple Integrals)

e.g. Evaluate $\int_0^3 \int_0^4 \int_{x=y/2}^{y/2+1} \left(\frac{2x-y}{z} + \frac{z}{3} \right) dx dy dz$. ($=: I$)

Sol. Let $u = \frac{2x-y}{z}$, $v = \frac{z}{3}$, $w = \frac{y}{z}$.

$$\Rightarrow x = u + w, \quad y = 2w, \quad z = 3v$$

$$\Rightarrow \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 3 & 0 \end{vmatrix} = -6 \Rightarrow \left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| = 6.$$

$$0 \leq z \leq 3 \Leftrightarrow 0 \leq v \leq 1$$

$$0 \leq y \leq 4 \Leftrightarrow 0 \leq w \leq 2$$

$$\frac{y}{2} \leq x \leq \frac{y}{2} + 1 \Leftrightarrow w \leq u + w \leq w + 1 \Leftrightarrow 0 \leq u \leq 1$$

$$\begin{aligned} \therefore I &= \int_0^1 \int_0^1 \int_0^2 (u+v) 6 \, dw \, dv \, du = 12 \int_0^1 \int_0^1 (u+v) \, dv \, du \\ &= 12 \int_0^1 \left(uv + \frac{1}{2} v^2 \right) \Big|_{v=0}^1 du = 12 \int_0^1 \left(u + \frac{1}{2} \right) du \\ &= 12 \left(\frac{1}{2} u^2 + \frac{1}{2} u \right) \Big|_{u=0}^1 = 12. \end{aligned}$$

Spherical coordinates

Consider finding the mass of a closed ball B with radius a , centered at $(0, 0, 0)$:

$$\text{mass}(B) = \iiint_B \rho(x,y,z) dV = \underbrace{\int_0^{2\pi} \int_0^a \int_{-\sqrt{a^2-x^2-y^2}}^{\sqrt{a^2-x^2-y^2}} \rho(x,y,z) dz r dr d\theta}_{\text{Cylindrical}}$$

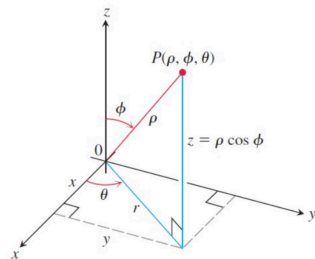
The inner integral is not friendly 😊.

Spherical coordinates may help.

Given any point P in the xyz -space, we can represent the point by (ρ, ϕ, θ) , where

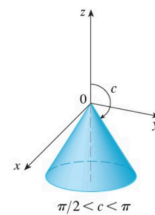
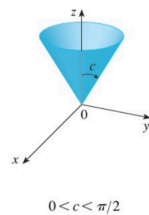
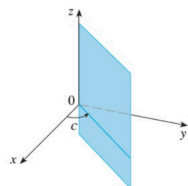
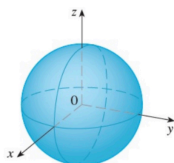
- ▶ ρ is the distance from P to the origin ($\rho \geq 0$);
- ▶ ϕ is the angle made from the positive z -axis to $\overrightarrow{OP}$ ($0 \leq \phi \leq \pi$), and;
- ▶ θ is the same as in cylindrical coordinates.

The coordinate system above in (ρ, ϕ, θ) is called the **spherical coordinate system**.



The figures below, from left to right, show the surfaces for:

1. $\rho = c$.
2. $\theta = c$.
3. $\phi = c$.



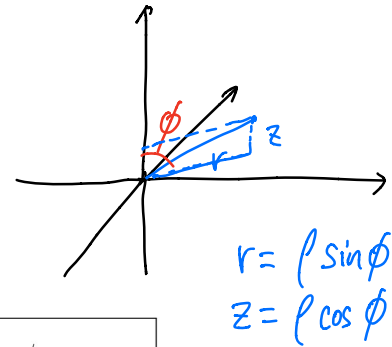
Conversion Between Coordinate Systems

Let r be as in cylindrical coordinates. Then

$$z = \rho \cos \phi \quad \text{and} \quad r = \rho \sin \phi,$$

from which we can deduce

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$



By definition, we also have (given x, y & z)

$$\rho = \sqrt{x^2 + y^2 + z^2}.$$

Then we can find ϕ
and then θ
as well.

e.g. (a) Find a spherical coordinate equation for the sphere

$$x^2 + y^2 + (z - 1)^2 = 1.$$

(b) Find a spherical coordinate equation for the cone

$$z = \sqrt{x^2 + y^2}.$$

Ans: (a) $\rho = 2 \cos \phi$
 $0 \leq \phi \leq \pi/2$.

(b) $\phi = \pi/4$.

Integration with Spherical Coordinates

When evaluating $\iiint_E f(x, y, z) dV$ with spherical substitutions

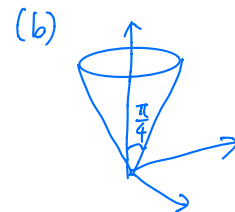
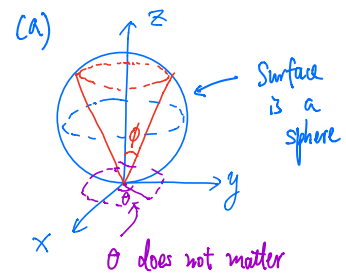
$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

the absolute value of the Jacobian is

$$\left| \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} \right| = \rho^2 \sin \phi,$$

so the change of differentials becomes

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta.$$



Check : $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$,

$$\frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)} = \begin{vmatrix} \sin \phi \cos \theta & \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\rho \sin \phi & 0 \end{vmatrix}$$

Expand along Row 3

$$= \cos \phi \begin{vmatrix} \rho \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \rho \cos \phi \sin \theta & \rho \sin \phi \cos \theta \end{vmatrix} - (-\rho \sin \phi) \begin{vmatrix} \sin \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \rho \sin \phi \cos \theta \end{vmatrix}$$

$$= \cos \phi (\rho^2 \cos \phi \sin \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta) + \rho \sin \phi (\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta)$$

$$= \rho^2 \sin \phi (\cos^2 \phi + \sin^2 \phi) = \rho^2 \sin \phi.$$

Since $0 \leq \phi \leq \pi$, $\left| \frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)} \right| = \rho^2 \sin \phi.$

Example

(a) Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV$, where B is the closed ball centered at the origin with radius 1.

(b) Find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$.

Ans:

(a) $\frac{4\pi}{3}(e-1).$

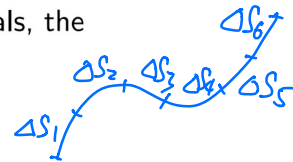
(b) $\frac{\pi}{8}.$

e.g. Find the volume of the solid E , where E is the intersection of $x^2 + y^2 \leq 1$ (E_1) and $x^2 + y^2 + z^2 \leq 4$ (E_2).

Line Integrals of Real-Valued Functions

Suppose we would like to compute the mass of a wire C in the space with linear density $f(x, y, z)$. By breaking C into smaller pieces and picking points (x_k, y_k, z_k) from the subintervals, the mass is approximately

$$\sum_{k=1}^n f(x_k, y_k, z_k) \Delta s_k.$$



This motivates the definition of line integrals.

Definition

If f is a real-valued ^{function} defined on a curve C , then the **line integral of f along C** ~~(with respect to arc length)~~ is

$$\int_C f(x, y, z) \, ds := \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(x_k, y_k, z_k) \Delta s_k,$$

s is arc length parameter
 P partition C into pieces

provided that the limit exists.

- For plane curve C , the line integral of $f(x, y)$ along C is $\int_C f(x, y) \, ds$, defined to be $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(x_k, y_k) \Delta s_k$.
- Line integrals can be defined similarly for an n -variable real-valued function $f(x_1, \dots, x_n)$ and curves C in $\mathbb{R}^n$.
- Will focus on $\mathbb{R}^2$ and $\mathbb{R}^3$.

Computation

- ▶ A function $\mathbf{r} : I \rightarrow \mathbb{R}^n$ is called **smooth** if $\mathbf{r}'$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$ for every t . A curve having a smooth parametrization is called a **smooth curve**. } Recall
- ▶ If C is given by a smooth parametrization

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad a \leq t \leq b,$$

and f is continuous on C , then $\int_C f \, ds$ exists, and

$$\begin{aligned} \int_C f(x, y, z) \, ds &= \int_a^b f(x(t), y(t), z(t)) |\mathbf{r}'(t)| \, dt \\ &= \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt. \end{aligned}$$

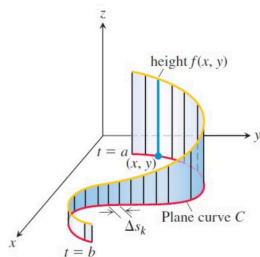
Speed of movement

- ▶ If $f \equiv 1$, then $\int_C f \, ds$ just gives the arc length of C .

- For curves in $\mathbb{R}^2$, it is similar:

$$\int_C f(x, y) \, ds = \int_a^b f(x(t), y(t)) |\mathbf{r}'(t)| \, dt = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| \, dt.$$

- ▶ The line integral of a nonnegative function $f(x, y)$ along a plane curve can be interpreted as the area of the "fence" between the graph and the xy -plane, as indicated below.



(At least) Two meanings of $\int_C f(x, y) \, ds$ for nonnegative f :

- Mass of a wire.
- Area of a fence/wall.

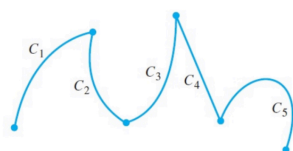
e.g. Evaluate $\int_C (2+x^2y) ds$, where C is the upper half of $x^2+y^2=1$.

Ans: $2\pi + \frac{2}{3}$.

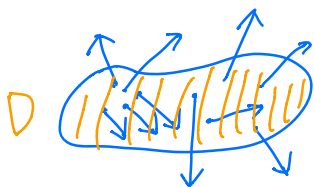
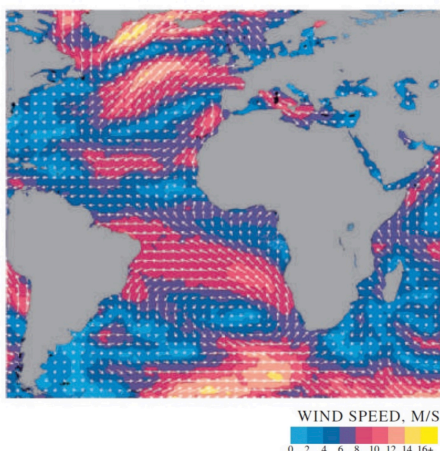
Remark

If C is a **piecewise smooth** curve, that is, if C is a union of finitely many smooth curves $C_1, C_2, \dots, C_n$, where the initial point of C_{i+1} is the terminal point of C_i , then

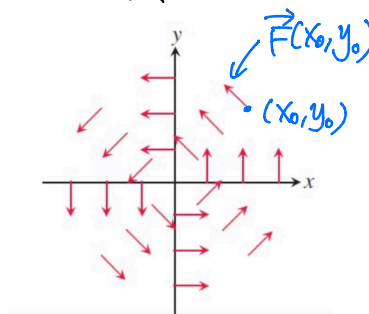
$$\int_C f ds = \int_{C_1} f ds + \int_{C_2} f ds + \dots + \int_{C_n} f ds.$$



Vector Fields



Def: Let $D \subseteq \mathbb{R}^n$ be a set of points. A **vector field** (in $\mathbb{R}^n$) is a function $\vec{F}: D \rightarrow \mathbb{R}^n$ that assigns each point in D a vector in $\mathbb{R}^n$.



Examples of vector fields:

- Velocity fields
- Force fields
- Gradient fields

We will focus on vector fields in $\mathbb{R}^2$ and $\mathbb{R}^3$, which we will write as

$$\mathbf{F}(x, y) = \langle M(x, y), N(x, y) \rangle$$

and

$$\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z) \rangle.$$

Line Integrals of Vector Fields

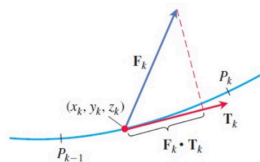
A Motivation: Work

A basic formula in physics states that

$$\text{Work} = \text{Force} \cdot \text{Distance},$$

if the force is a constant. Suppose that a force field $\mathbf{F}$ in the space drags a particle along a smooth curve C . To approximate the total work done by $\mathbf{F}$ in moving the particle, we may do the following.

- Partition C into $\{P_0, P_1, \dots, P_n\}$.
- The work done by $\mathbf{F}$ in moving the particle from P_{k-1} to P_k is approximately $\mathbf{F}(x_k, y_k, z_k) \cdot \mathbf{T}(x_k, y_k, z_k) \Delta s_k$, as shown in the following figure.



- We expect the total work done to be

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \mathbf{F}(x_k, y_k, z_k) \cdot \mathbf{T}(x_k, y_k, z_k) \Delta s_k.$$

We define the **work** done by a continuous force field $\mathbf{F}$ in moving a particle along C to be

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds.$$

This is an example of a line integral of a vector field.

Definition

Let $\mathbf{F}$ be a continuous vector field defined on a smooth curve C .

The **line integral of $\mathbf{F}$ along C** is

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds.$$

Note that
 $f(x, y, z) := \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z)$
 is a real-valued function, so
 $\int_C \mathbf{F} \cdot \mathbf{T} \, ds = \int_C f \, ds.$

*talked about this
in previous section*

Remark

If C is parametrized by $\mathbf{r}(t)$, for $t \in [a, b]$, then

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} |\mathbf{r}'(t)| \, dt = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt.$$

For this reason, the line integral $\int_C \mathbf{F} \cdot \mathbf{T} \, ds$ is often also written as

$$\int_C \mathbf{F} \cdot d\mathbf{r}.$$

Alternatively

$$\int_C \vec{F} \cdot \vec{T} \, ds$$

$$= \int_C \vec{F} \cdot \frac{d\vec{r}}{ds} \, ds$$

$$= \int_C \vec{F} \cdot d\vec{r}$$